higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 4

(12020524)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: GENERAL

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 The process during which hydration takes place is called curing.

1.1.2 Shuttering is when we use plastering to shape wet concrete into a specific form.

1.1.3 When placing the formwork, it is not common practice to use releasing agent.

1.1.4 A screed is a mixture of cement and sand placed on a concrete slab to provide a smooth flat finish.

1.1.5 We use a slump test to test the consistency of brick.

(5 × 1) (5)

1.2 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 ... is the surface to which the plaster bonds.

A Substrate
B Plaster
C Floating
D Grout

1.2.2 To ... is to add water to mortar which is starting to stiffen.

A top
B retemper
C mix
D flush

1.2.3 We use a BRE test once the screeds are ... days old.

A 28
B 30
C 7
D 14

1.2.4 ... is the process that causes water to accumulate on the top surface of the fresh concrete and the presence of this water layer may impair adhesion unless suitably dealt with.

A Curing
B Debonding
C Bleeding
D Scoring

1.2.5 A ... joint is used on construction projects to isolate or limit cracking caused by internal or external stresses on the structure.

A control
B cut
C rake
D vee

(5 × 1) (5)
[10]

QUESTION 2: PLASTERING

2.1 For a good bond, the prepared surface should be rough, hard and clean. Re-arrange the following steps in the proper sequence of plastering in the ANSWER BOOK.

2.1.1 Make sure that the wall is strong enough.

2.1.2 Test the area for water absorbency.

2.1.3 Check that the substrate is plane and vertical.

2.1.4 Remove all protrusions.

2.1.5 Remove all dust, dirt and oily substances from the walls.

2.1.6 On the day before plastering, wet the area well.

2.1.7 Clear the surrounding area of all the materials and tools that you do not need for the job.

2.1.8 If necessary, roughen the substrate to allow the plaster to bond properly with it.

(8 × 1) (8)

2.2 Name THREE tools that you require for plastering. (3)

2.3 Briefly discuss how to prepare masonry substrate by raking the joints before commencing with plastering. (8)

2.4 Explain FIVE steps you can use to prepare off-shutter concrete surfaces for plastering. (5)

2.5 How thick should plaster be if only a single coat is to be applied? (2)

2.6 Name THREE places where you should cut joints through plaster. (3)

2.7 What is off-shutter concrete? (1)
[30]

QUESTION 3: SCREEDS AND TOPPINGS

3.1 Explain why it is important to check the concrete slab for readiness before applying the screed/topping. (3)

3.2 Explain THREE types of screeds and toppings. (6)

3.3 Why do we float the surface? (4)

3.4 Why are screed battens placed on the concrete substrate before applying the screed/topping? (1)

3.5 What is a colour hardener, and why do we sometimes use it when finishing a topping or screed? (4)

3.6 To what depth should you cut the joints along the perimeter wall and around columns? (1)

3.7 What do you use to help remove grease from the substrate? (2)

3.8 Explain the FIVE steps in applying the slush coat. (5)

3.9 What is an intercoat? (2)

3.10 Name TWO types of cement you should use to mix screeds and toppings. (2)
[30]

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QUESTION 4: TILLING

4.1 Where should the following floor coverings be used:

4.1.1 Granolithic floors

4.1.2 PVC tiles

4.1.3 Thermoplastic tiles

(3 × 1) (3)

4.2 What would happen if you laid tiles against a gypsum plaster-board backing? (4)

4.3 Draw to a suitable scale the front view and top view of a grouted tiled square plastered column.

Specifications:

- Tiles 150 mm × 150 mm
- Column 400 mm × 400 mm
- Only FOUR courses of tiles to be shown on the elevation. (13)

4.4 Briefly explain how to check the tolerance of a finished tile floor. (5)

4.5 Describe how to apply grout to fill the joints between tiles. (4)

4.6 How do you check that the adhesive is evenly spread over the underside of the tile? (1)

[30]

TOTAL: 100

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